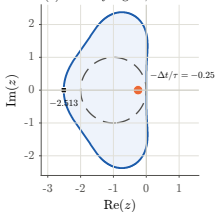
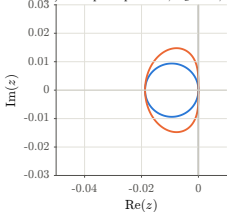


(a) Stability region, $z = \lambda\Delta t$



(b) Transport spectrum, $N_x = 50,000$



— SSP-RK3 $|R(z)| \leq 1$
- - Forward Euler

● BGK mode (solver)
■ Real-axis limit

— Upwind, $\nu = 0.0094$
— MUSCL (unlimited), $\nu = 0.0094$

— SSP-RK3 boundary